What do I do with a list of genes?

Making sense of data

BMMB 852: Applied Bioinformatics

So you have a list of genes

Now what?

Results produced by algorithms are rarely immediately meaningful

Interpreting the results

is a distinct and separate process.

Being good at data interpretation is an extremely valuable skill

Many bioinformaticians focuson exessively on algorithms, tools etc.

Making sense of data sometimes becomes an afterthought!

Don't be one of them.

What your results will look like

A surprising number of "final" data analysis results will be on of:

1. A **list of names**
2. A **list of names** with a **single value**
3. A **list of names** with a **matrix of values**

The names may be **gene**, **transcript** or other **feature names**.

The question becomes to how do you interpret the list?

A list of genes

See the online lecture information for `gene-list.txt`, `gene-values.txt` and `gene-matrix.txt`

```
cat gene-list.txt | head -5
```

prints:

```
ACE
ADRB2
ADRB3
AGRP
AKR1C2
```

The list is limited to a *small subset* of all names!

A list of genes with values

Displays a name with a single measurement:

```
cat gene-values.txt | head -5
```

prints:

```
gene        value
Tmem132a    1.04E-12
Myl3        6.67E-14
Myl4        3.27E-09
Hspb7       1.27E-07
```

One "value" for each row.

A gene matrix

Displays a name with multiple measurements

```
cat gene-matrix.txt | head -5
```

prints

```
gene      C1      C2       C3       M1      M2      M3
Tmem132a  7349.5  10604.4  11400.6  694.7   709.3   760.2
Myl3      1207.1  1345.0   1247.6   2222.9  3041.3  2819.0
Myl4      2468.6  2588.5   2840.4   3963.2  5044.9  4824.7
Hspb7     562.5   610.3    647.0    947.0   1300.0  1144.7
```

Multiple "values" for each row.

But what does it all mean?

2002, Donald Rumsfeld (Secretary of Defence):

On the Iraq war:

- **There are known knowns**. There are things we know that we know.
- **There are known unknowns**. That is to say, there are things that we now know we don't know.
- **But there are also unknown unknowns**. There are things we do not know we don't know

Much ridiculed back then - yet quite deep and thoughtful - especially regarding scientific inquiry

COMMENTARY

Known knowns, known unknowns, unknown unknowns and the propagation of scientific enquiry

FREE

David C. Logan

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Issue Section: eXtra Botany

In February 2002, Donald Rumsfeld, the then US Secretary of State for Defence, stated at a Defence Department briefing: 'There are known knowns. There are things we know that we know. There are known unknowns. That is to say, there are things that we now know we don't know. But there are also unknown unknowns. There are things we do not know we don't know.' As a result, he was almost universally lampooned since many people initially thought the statement was nonsense. However, careful examination of the statement reveals that it does make sense, indeed the concept of the unknown unknown existed long before Donald Rumsfeld gave it a new audience.

Is this a complete enumeration?

1. Known knowns
2. Known unknowns
3. Unknown unknowns

Is there a fourth category?

Unkown knows!

The things we don't know that we know.

"There are known knowns"

First you ought to understand:

---

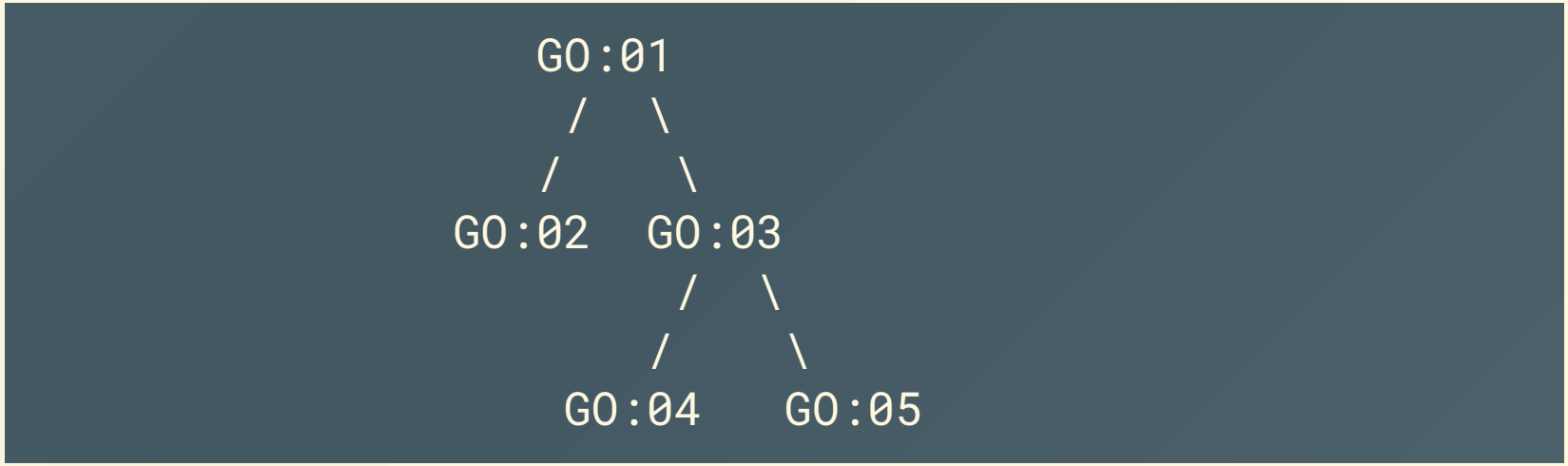
1. How is the current knowledge represented.
2. How is the current knowledge searched.

Every method will search your data against the known knowns.

Recap: The Gene Ontology

What is the GO?

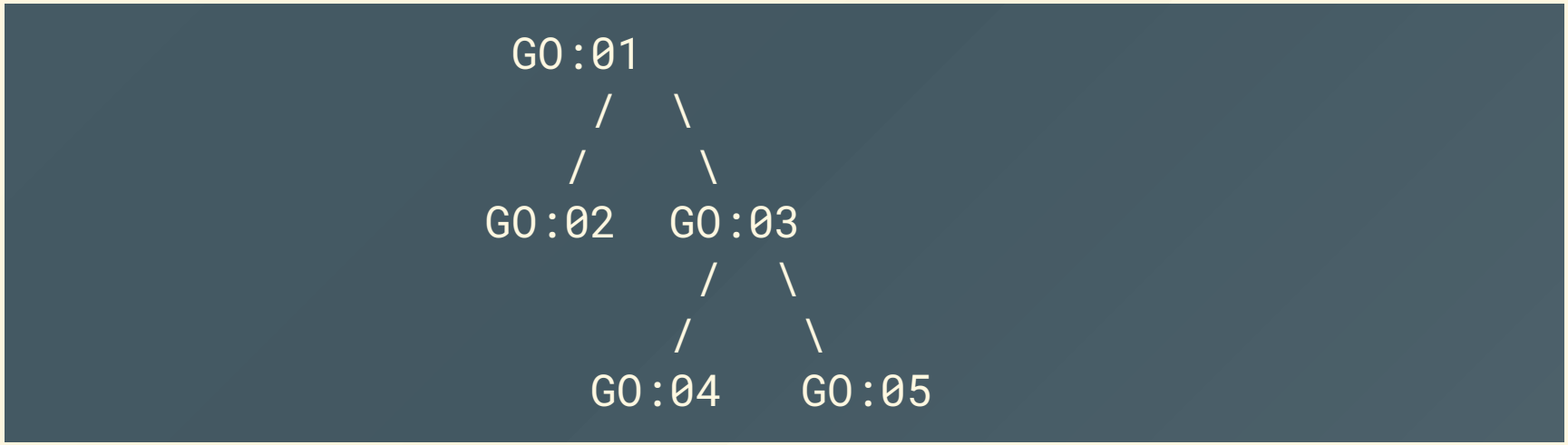
- Definition of GO Terms layed out as a tree:



- And an association of gene products with terms.

```
Gene  A     GO:05
```

Multiple types of edges



- `is_a` - a subtype
- `part_of` - complete containment
- `has_par` - partial containment
- `regulates` - partial effect

Direct and indirect links



Everything labeled as `GO:05` will also "be" a `GO:03` and `GO:01`

A gene known to be `GO:03` could also be `GO:04` or `GO:05`, perhaps both, or perhaps neither.

Leaf nodes - nodes with no children

These are the most detailed representation.

```
GO:0008150 -> biological process
  GO:0032502 -> development process
    GO:0048856 -> anatomical structure development
      GO:0048513 -> animal organ developmen
        GO:0007507 -> heart development
          GO:0007512 -> adult heart development
```

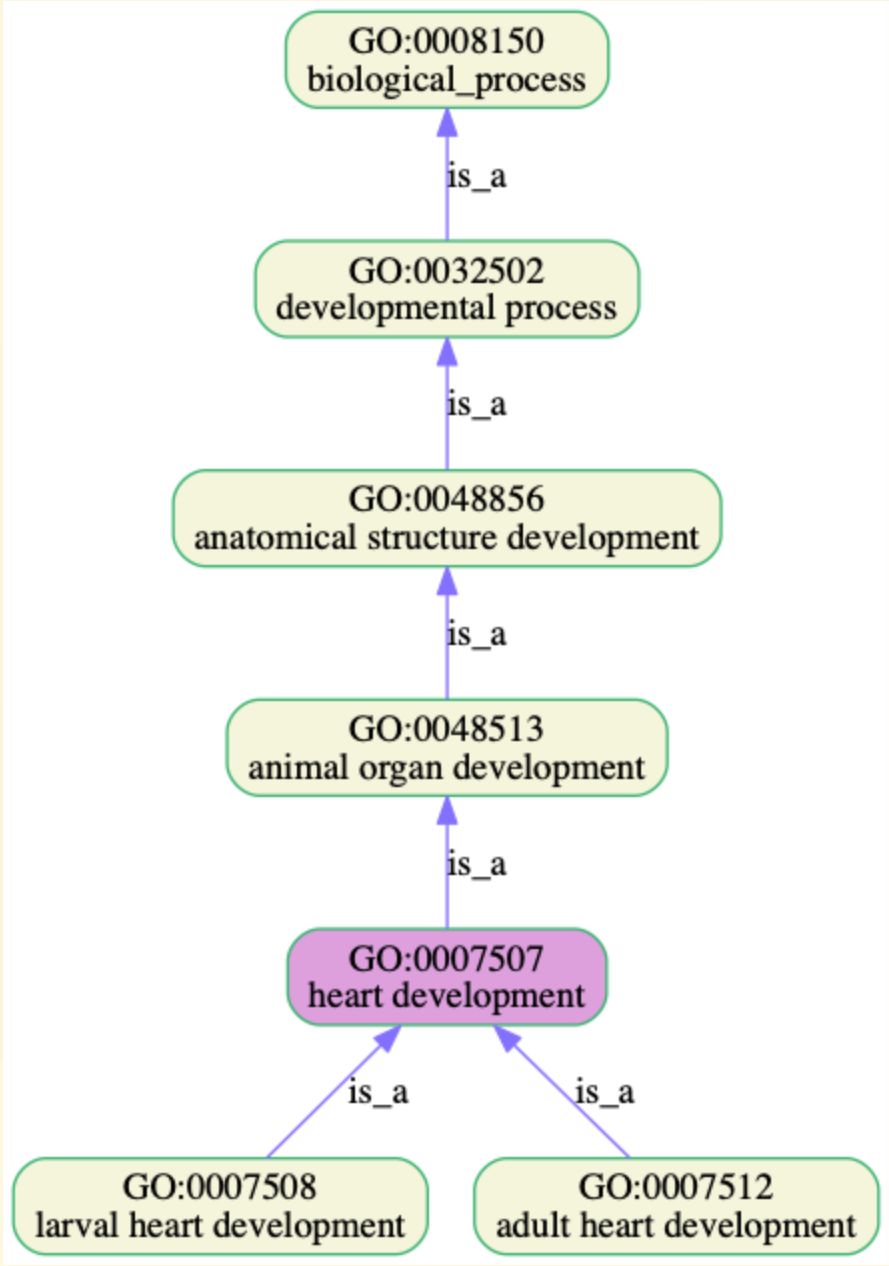
GO analysis from command line

Install with:

```
# Install software
conda install goatools

# Get the go.obo file
wget http://geneontology.org/ontology/go-basic.obo

# Visualize a term
plot_go_term.py --term GO:0007507 go.obo
```


GO:0008150
biological_process
is_a
GO:0032502
developmental process
is_a
GO:0048856
anatomical structure development
is_a
GO:0048513
animal organ development
is_a
GO:0007507
heart development
is_a
is_a
GO:0007508
larval heart development
GO:0007512
adult heart development

See what this gives you

```
plot_go_term.py --term GO:0000001
```

Tacit limitations

Makes use of only the `is_a` relationships between concepts.

Does not use make the `part_of` or other relationships.

It is very typical of bioinformatics tools to not be explicit of their limitations.

Investigate the GO data to understand it

The gene ontology definition file (detailed commands in the book)

```
wget http://purl.obolibrary.org/obo/go-basic.obo
```

how many lines:

```
cat go.obo | wc -l
# 632140
```

Page through it with

```
more go-basic.obo
```

GO Terms

The content of the core `go-basic.obo` file is constructed
of records in the form:

```
[Term]
id: GO:0000002
name: mitochondrial genome maintenance
namespace: biological_process
def: "The maintenance of the structure and integrity of
  the mitochondrial genome; includes replication and
  segregation of the mitochondrial chromosome." [GOC:ai, G
is_a: GO:0007005 ! mitochondrion organization
```

The `is_a` line indicates the *parent* of the term.
If your "concept" is not in this file tools will not find

Manipulate GO files from command line

Once you identify the patterns in the records, then you can search for various content:

```
cat go.obo | grep "namespace: biological_process" | wc -l
# 30583

cat go.obo | grep "namespace: molecular_function" | wc -l
# 12123

cat go.obo | grep "namespace: cellular_component" | wc -l
# 4300
```

Every functional enrichment tool uses this file as its basis.

Search the file for functions

You can also get previous and following lines by passing the `-B` (before) `-A` (after) options to `grep`.

```
cat go.obo | grep "lactase activity" -B 2 -A 5 | head -8
```

Prints:

```
[Term]
id: GO:0000016
name: lactase activity
namespace: molecular_function
def: "Catalysis of the reaction: lactose + H2O = D-glucose
synonym: "lactase-phlorizin hydrolase activity" BROAD [EC:
synonym: "lactose galactohydrolase activity" EXACT [EC:3.2
xref: EC:3.2.1.108
```

The association file for Homo Sapiens

From the GO download page, copy the link then:

```
wget http://geneontology.org/gene-associations/goa_human.g

# Unzip the compressed file.
gunzip goa_human.gaf.gz
```

How big is the resulting file:

```
cat goa_human.gaf | wc -l
# 425901
```

There you have it. `425,901` known functions for the human genes.

What is in the association file?

There is a readme with the file (on the web) and you can download that the same way.

You can also page through the file

```
cat goa_human.gaf | more
```

Comments are specfied with `!` the rest are tab separated and column oriented data.

Remove the lines starting with `!` to simplify it.

```
cat goa_human.gaf | grep -v '!' > assoc.txt
```

What properties do the data have?

The GAF format states that column 3 has to be a

> *a symbol that means something to a biologist wherever possible (a gene symbol, for example)*

```
cat assoc.txt | cut -f 3 | head
```

Prints

```
DNAJC25-GNG10
DNAJC25-GNG10
DNAJC25-GNG10
HDGFRP3
```

How many gene symbols?

```
cat assoc.txt | cut -f 3 | sort | uniq -c | wc -l
# 19421
```

Most genes appear to have at least one entry. `425,901` over `19,421` genes means on average about `22` annotation per gene.

But the annotations are not evenly distributed.

Annotation distribution

Redirect output into `gene_counts.txt`:

```
cat assoc.txt | cut -f 3 | sort | uniq -c | sort -k1,1nr >
cat gene_counts.txt | head
```

The "top" genes have annotations way above the `22`

```
724 TP53
669 GRB2
637 EGFR
637 UBC
580 RPS27A
570 UBB
565 UBA52
511 CTNNB1
422 SRC
```

Command line analytics with datamash

A handy tool called `datamash` lets you do data analytics at command line.

```
# Activate your enviroment
source activate bioinfo

# Get help on datamash
datamash --help
```

Unfortunately the `uniq -c` command pads numbers with a variable number of spaces. We need to `squeeze` those into a single space. `tr -s` can do that.

```
cat gene_counts.txt | tr -s ' '
```

Working with datamash

Average number annotations:

```
cat gene_counts.txt | tr -s ' ' | datamash -t ' ' mean 2
# 21.928170537048
```

You can list multiple operations at a time:

```
cat gene_counts.txt | tr -s ' ' | datamash -t ' ' mean 2
# 21.928170537048 1 724
```

Functional analysis

Typically described with "overloaded" words, roughly the following categories:

- Overrepresentation analysis (ORA)
- Gene set enrichment analysis (GSEA)
- Pathway analysis

The definitions are a bit nebulous and overlapping.

Overrepresentation

Are known biological functions over-represented (enriched) in an list relative to an expectation.

For example:

- Out of my list of **20** genes I see **10** annotated as **heart development (GO:0007507)**. Does that mean this function is enriched?

A typical overrepresentation study

```
# How many genes are annotated in total
cat goa_human.gaf | cut -f 3 | sort | uniq -c | wc -l
19675

# How many genes annotated in the subtree of GO:0007507
cat goa_human.gaf | grep GO:0007507 | wc -l
207
cat goa_human.gaf | grep GO:0007508 | wc -l
0
cat goa_human.gaf | grep GO:0007512 | wc -l
16
```

Thus we expect to see **223** out of **19675** (1%) but we see **10** out of **20** (50%). Is heart development overrepresented?

List some potential pitfalls

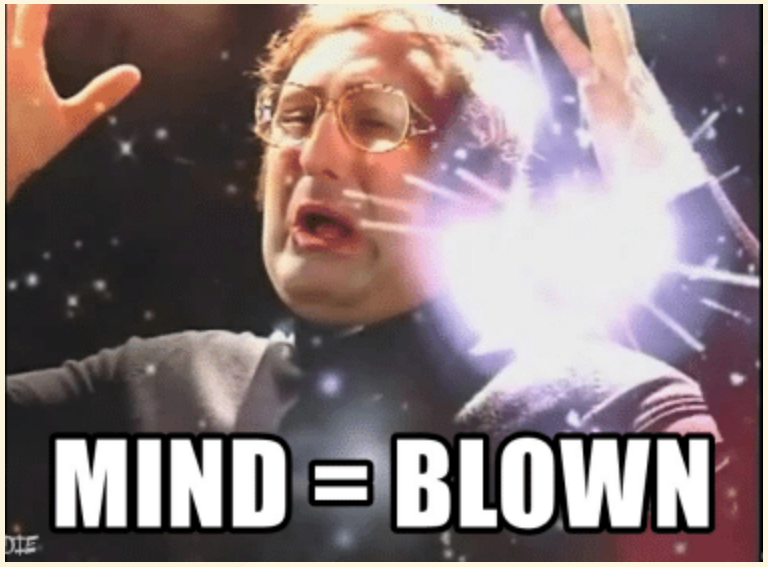
Many corrections are needed

1. Multiple testing
2. Functional categories are not independent (GO is a tree!)
3. The unknown unknowns

Every publication and tool claims to have addresses one or more weaknesses.

But there is no "gold" standard to evaluate against! Why?

We don't know when we are correct



The unknown known!

There is a whole cottage industry of functional tools.

Never clear which one works "better" than others

People pick the one that seems to work for them - take that SCIENCE!

Gene set enrichment analysis

Typically

1. Requires a gene list with values
2. All genes/transcripts should be present
3. The values should allow a ranking of the rows (high/low)

The question it answers:

Are there lists of functions common to genes listed in the top/bottom

The main difference

- Overrepresentation: the subset is already known -> get functions
- Gene set enrichment: a ranking is known -> get functions that are enriched in the top/bottom

Analysis tools

The book shows several options (many more exists):

- **g:Profiler**
- **Panther**
- **DAVID**
- **ermineJ**

Tools come with different tradeoffs and my better suited for different problem sets.

It is not clear beforehand which tool works for a given problem.

Will different tools produce different results?

Biostar Quote of the Day: Why does each GO enrichment method give different results?

> *I'm new to GO terms. In the beginning it was fun, as long as I stuck to one algorithm. But then I found that there are many out there, each with its own advantages and caveats (the quality of graphic representation, for instance). [...] As a biologist, what should I trust? Deciding on this or that algorithm may change the whole story!*

Data selection for the lecture

If you are pursuing a research project already you may have your own data available. Use that when you follow the examples in the book.

You can also make your own data. Example: take the top 20 most annotated genes from GO and see what is common about them.

You may also download the `gene-list.txt` from the lecture website.

The same applies to the homework.